

Stakeholder Engagement Plan

Optimizing RAG Document Retrieval for Agronomic Advice

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Selected Project	Optimizing RAG Document Retrieval for Agronomic Advice
Challenge	Challenge 4 — Stakeholder Engagement Plan
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1. Stakeholder Identification

Stakeholder Group A: Smallholder Farmers

Who they are: Smallholder farmers who grow staple and cash crops on small plots of land, and who would be the direct end users asking questions of the retrieval system (either directly, or through an extension worker on their behalf). **Why they are important:** They are the intended primary beneficiaries of the project — the entire premise of the system is to get them faster, more relevant agronomic guidance. If the system does not reflect how they actually ask questions or what problems they actually face, it fails at its core purpose regardless of technical performance. **What matters most to them:** Practical, trustworthy, locally relevant answers they can act on quickly; being understood in their own words (including local terms for crops and problems) rather than having to translate their question into technical language; and not being given advice that risks their harvest if wrong.

Stakeholder Group B: Agricultural Extension Officers / Organisations

Who they are: Government and NGO agricultural extension officers and the organisations that employ them, who currently serve as the main trusted intermediary between agronomic knowledge and farmers. **Why they are important:** They understand, better than our team does, which questions farmers actually bring to them, which existing reference material is trustworthy versus outdated, and how a retrieval tool would realistically fit into (or disrupt) their existing workflow. They are also a channel through which the tool could reach farmers responsibly, with human oversight. **What matters most to them:** That the tool supports rather than undermines their role and professional authority; that it draws on material they themselves would consider credible; and that it does not introduce liability if it surfaces poor advice under their organisation's name.

2. Potential Benefits and Risks

Smallholder Farmers

Benefits: Faster access to relevant agronomic guidance; reduced dependence on the availability of an extension officer for simple, well-documented questions. **Risks/concerns:** Being given a confidently-presented but poorly-matched document for their specific situation or region; a system that works well for common crops/questions but poorly for local or under-documented ones, creating unequal benefit.

Ethical/social/cultural/economic considerations: Risk of eroding trust in extension services if the tool is seen as a replacement rather than a complement; risk of excluding farmers with lower literacy or without access to a smartphone/internet from ever benefiting.

Agricultural Extension Officers / Organisations

Benefits: A tool that helps them locate the right reference material faster, freeing time for higher-value, judgement-based interactions with farmers. **Risks/concerns:** Feeling that the tool positions AI as a replacement for their expertise rather than a support to it; concern about being held responsible for advice sourced through a tool they did not build or fully vet. **Ethical/social/economic considerations:** Potential job-role anxiety if the tool is perceived as a step toward automating extension work; need for clear boundaries on where human judgement remains essential (e.g. anything safety-critical or highly context-dependent).

3. Engagement Method

Smallholder Farmers — Participatory Design Sessions

Why appropriate: Participatory Design directly involves the people most affected by a system in shaping it, rather than designing on their behalf based on assumptions. Given that our team is not made up of practising farmers, this method is essential to surface how farmers actually phrase questions, which local/colloquial crop and problem names we should account for, and what "a good answer" looks like from their perspective. **Why suitable for this group:** It is low-barrier and concrete — farmers can react to real example questions and real example documents rather than needing to understand retrieval systems abstractly. **How it would be conducted:** In partnership with a local extension office, we would run a small number of short, informal sessions (ideally in the local language, with an extension officer or team member translating as needed) where farmers are shown sample farmer questions and asked: is this how you'd actually ask it? Is this the document you'd find useful? What's missing? Sessions would be kept short and low-pressure to respect farmers' time.

Extension Officers / Organisations — Deliberative Workshop

Why appropriate: A Deliberative Workshop brings a small group of extension officers together to discuss and weigh trade-offs collectively (e.g. how the tool should be positioned relative to their own role, what quality bar a source document needs to meet, where human sign-off should remain mandatory), rather than collecting isolated individual opinions. **Why suitable for this group:** Extension officers already have professional standards and institutional context to draw on, so a structured group discussion lets them collectively reason about risk and appropriate use in a way a simple survey would not capture. **How it would be conducted:** A single half-day workshop with a small group of officers from partner organisations, structured around three questions: which source documents do they already trust and use; where would they want a mandatory human review step before advice reaches a farmer; and how should the tool be introduced so it reads as support rather than replacement.

4. Influence on Project Design

Feedback from these two engagements would concretely shape multiple parts of the project. **Dataset:** Farmer sessions would surface local/colloquial terms and commonly-asked question phrasings we should test our retrieval against, and may reveal crop or problem categories missing from our current document coverage. **Model design:** Extension officer input on where human review should be mandatory could lead us to add a confidence threshold or flag low-similarity results for human sign-off rather than presenting every result with equal confidence. **Outputs:** Farmer feedback on what "a useful answer" looks like may lead us to present retrieved documents with a short plain-language summary of relevance rather than the raw document alone. **Evaluation:** Officer-identified "trusted vs. not trusted" source documents would let us weight our relevance judgements by source credibility, not just topical match. **Deployment decisions:** Both groups' concerns about

the tool replacing rather than supporting extension officers would push us toward positioning and piloting the tool as an officer-facing aid first, rather than a direct-to-farmer app, until trust and accuracy are further established.

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